

# **Week 23: Project Implementation – Frontend**

## ***1. All Pages & Components***

Frontend development included building all required pages and reusable UI components. Pages such as home, login, dashboard, and profile were implemented. Component-based design improved code organization and reusability.

## ***2. State Management***

State management was used to handle application-wide and component-level data efficiently. Tools like React state, Context API, or Redux were applied to manage dynamic updates and user interactions. This improved application consistency.

## ***3. API Integration***

Frontend components were connected with backend APIs using Axios or Fetch API. This enabled real-time data retrieval, submission, and updates. API integration is essential for full-stack communication.

## ***4. Responsive UI***

Responsive UI design ensured the application worked smoothly on desktops, tablets, and mobile devices. CSS, Flexbox, Grid, and media queries improved adaptability. This enhanced user experience across different screen sizes.

## ***Conclusion***

This week focused on frontend implementation. By building pages, managing state, integrating APIs, and designing responsive interfaces, a complete user-friendly frontend was developed.